Single variable - mixed practice

Mix linear and quadratic substitution in one short practice session.

Name: _____ Suggested time: 10 minutes

Show your working. Use brackets when substituting negative values.

01 Given x=6

Evaluate: $7x+10$

02 Given x=-3

Evaluate: x^2+4x-8

03 Given x=6

Evaluate: $7x+2$

04 Given x=0

Evaluate: x^2+3x+4

05 Given x=1

Evaluate: $6x-8$

06 Given x=5

Evaluate: x^2+3x-4

07 Given x=8

Evaluate: $6x-8$

08 Given x=-3

Evaluate: x^2+6x+1

Answer sheet

Single variable - mixed practice | 8 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 52

02 -11

03 44

04 4

05 -2

06 36

07 40

08 -8

Worked solutions

Single variable - mixed practice

01 Given x=6

$7x+10$

Substitute x=6.

$7\times 6 +10$

$42 +10$

52

02 Given x=-3

x^2+4x-8

Substitute x=-3.

$(-3)^2 +4\times (-3) -8$

$9 -12 -8$

-11

03 Given x=6

$7x+2$

Substitute x=6.

$7\times 6 +2$

$42 +2$

44

04 Given x=0

x^2+3x+4

Substitute x=0.

$(0)^2 +3\times 0 +4$

$0 +4$

4

Worked solutions

Single variable - mixed practice

05 Given x=1

$6x-8$

Substitute x=1.

$6\times 1 -8$

$6 -8$

-2

06 Given x=5

x^2+3x-4

Substitute x=5.

$(5)^2 +3\times 5 -4$

$25 +15 -4$

36

07 Given x=8

$6x-8$

Substitute x=8.

$6\times 8 -8$

$48 -8$

40

08 Given x=-3

x^2+6x+1

Substitute x=-3.

$(-3)^2 +6\times (-3) +1$

$9 -18 +1$

-8